

4.9. Message Privacy

Privacy processing of a message describes the obfuscation and deobfuscation of the message header fields after encryption and before decryption.

The detailed steps involved in privacy processing of outgoing and incoming Matter messages are described in [Section 4.9.3, “Privacy Processing of Outgoing Messages”](#) and [Section 4.9.4, “Privacy Processing of Incoming Messages”](#) respectively. They rely on the cryptographic primitives in [Section 3.7, “Message privacy”](#).

4.9.1. Privacy Key

The Privacy Key is a symmetric key specifically used for [Privacy Processing](#) that is derived from the [EncryptionKey](#) used for [Security Processing](#) of a given message. Given a Session ID reference to a specific Encryption Key, the Privacy Key is derived as follows:

```
PrivacyKey =
    Crypto_KDF
    (
        InputKey = EncryptionKey,
        Salt = [],
        Info = "PrivacyKey",
        Length = CRYPTO_SYMMETRIC_KEY_LENGTH_BITS
    )
```

4.9.2. Privacy Nonce

The Privacy Nonce is a nonce specifically used for [Privacy Processing](#) that is derived from the [SessionID](#) and [MIC](#) of the message. The Privacy Nonce SHALL be the [CRYPTO_AEAD_NONCE_LENGTH_BYTES](#)-octet string constructed as the 16-bit Session ID (in big-endian format) concatenated with the lower 11 (i.e. [CRYPTO_AEAD_MIC_LENGTH_BYTES-5](#)) bytes of the MIC:

```
PrivacyNonce = Session ID || MIC[5..15]
```

For example if Session ID is 42 (i.e. 0x002A) and the computed MIC is c5:a0:06:3a:d5:d2:51:81:91:40:0d:d6:8c:5c:16:3b:

```
Session ID = 00:2a
MIC = c5:a0:06:3a:d5:d2:51:81:91:40:0d:d6:8c:5c:16:3b
MIC[5..15] = d2:51:81:91:40:0d:d6:8c:5c:16:3b
PrivacyNonce = SessionID || MIC[5..15] = 00:2a || d2:51:81:91:40:0d:d6:8c:5c:16:3b
PrivacyNonce = 00:2a:d2:51:81:91:40:0d:d6:8c:5c:16:3b
```